SURJECTIVE FINITE MAPS OVER A QUASI-COMPACT OPEN OF A QCQS SCHEME EXTEND OVER THE WHOLE BASE BY ADDING THE BOUNDARY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13704-13711 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13713-13723, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Sister of the integral version collected this same wave (L16127) and of the fppf-domination duo (L16107/L16118); the finite case needs ZMT while the integral case only needs normalization.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma quasi-finite-separated-pass-through-finite (the ZMT compactification); elementary facts on scheme-theoretic images of quasi-compact opens and closed immersions which are dense; finiteness of coproducts of finite S-schemes.

Source reference. Original lemma 05K0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12457–12460. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. If $U \subset S$ is a quasi-compact open and $V \to U$ is a surjective finite morphism, then there exists a surjective finite morphism $\overline{V} \to S$ with $\overline{V} \times_S U$ isomorphic to V as schemes over U.*

Proof. By Zariski's Main Theorem (Lemma 05K0) we can assume V is a quasi-compact open in a scheme V' finite over S. After replacing V' by the scheme theoretic image of V we may assume that V is dense in V'. It follows that $V' \times_S U = V$ because $V \to V' \times_S U$ is closed as V is finite over U. Let Z be the reduced induced scheme structure on $S \setminus U$. Then $\overline{V} = V' \amalg Z$ works. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.